Chapter 3: Two-Dimensional Kinematics

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PHY201

Lecture 3

Part I

Vectors and Two-Dimensional Motion

Outline of Part I: Vectors and Two-Dimensional Motion

Introduction to Two-Dimensional Kinematics

In Chapter 2, we studied one-dimensional motion along a straight line.

In this chapter, we extend those ideas to two dimensions (motion in a plane).

- Examples of 2-D motion include a ball thrown at an angle, a car turning a corner, and a river current carrying a boat.
- Horizontal motion and vertical motion are independent of each other.
- $\therefore$ our strategy is to turn one 2-D kinematics problem into two 1-D kinematics problems.

Independence of Perpendicular Motions

The horizontal and vertical components of motion do not affect each other.

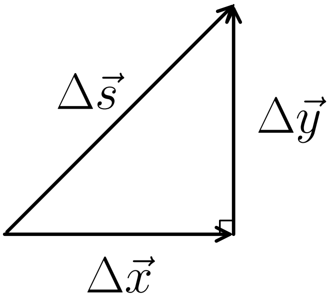
Falling Baseballs Experiment

If one baseball is dropped from rest and another is thrown horizontally at the same time from the same height:

- Both reach the ground at the same time.
- This demonstrates that horizontal motion does not affect vertical motion.

- Gravity acts in the y direction.
- y acceleration (gravity) only affects y velocity; it has no effect on x velocity.
- We analyze each direction separately when solving a 2-D kinematics problem.

Displacement in Two Dimensions



In 2-D, displacement has both a horizontal (x) and a vertical (y) component.

The magnitude of the resultant displacement from two perpendicular legs is found using the Pythagorean theorem:

$$s = \sqrt{(\Delta x)^2 + (\Delta y)^2}$$

where $\Delta\vec{x}$ is the horizontal displacement, $\Delta\vec{y}$ is the vertical displacement, and $\vec{s}$ is the resultant displacement.

Example: Displacement in Two Dimensions

Walking 9 blocks east and 5 blocks north gives a "distance traveled" of 14 blocks, but a displacement of:

$$s = \sqrt{9^2 + 5^2} = \sqrt{81 + 25} = \sqrt{106} \approx \boxed{10.3\ \text{blocks}}$$

Outline of Part I: Vectors and Two-Dimensional Motion

Vectors v. Scalars — Review

Definitions

- **Vector:** Quantity with both magnitude and direction
- **Scalar:** Quantity with magnitude only (no direction)

Vectors

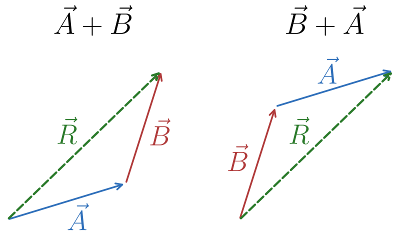
- Displacement $\Delta\vec{r}$
- Velocity $\vec{v}$
- Acceleration $\vec{a}$
- Force $\vec{F}$

Scalars

- Distance d
- Speed s
- Mass m
- Temperature T

Vectors are represented by arrows: length $\propto$ magnitude; orientation $=$ direction.

Graphical Vector Addition — Head-to-Tail Method



Commutative property:

$$\vec{A}+\vec{B}=\vec{B}+\vec{A}$$

The order of addition does not change the resultant.

Head-to-Tail Method

To add vectors $\vec{A}$ and $\vec{B}$ graphically:

1. Draw $\vec{A}$ to scale with a ruler and protractor.
2. Place the **tail** of $\vec{B}$ at the **head** of $\vec{A}$.
3. The resultant $\vec{R}=\vec{A}+\vec{B}$ runs from the tail of $\vec{A}$ to the head of $\vec{B}$.
4. Measure the magnitude and direction of $\vec{R}$.

Vector Subtraction

Vector Subtraction

The negative of a vector $\vec{B}$, written $-\vec{B}$, has the same magnitude but the opposite direction. Subtraction is defined as:

$$\vec{A} - \vec{B} = \vec{A} + \left(-\vec{B}\right)$$

Scalar Multiplication

If vector $\vec{A}$ is multiplied by scalar c:

- Magnitude becomes $\left|c \cdot \vec{A}\right|$
- Direction of $\vec{A}$ is unchanged if $c > 0$, but reversed if $c < 0$.

Check Your Understanding

You walk 3 blocks north and then 4 blocks east.

(a) What total distance did you travel?

(b) What is the magnitude of your displacement?

Come to lecture to see the answer.

Outline of Part I: Vectors and Two-Dimensional Motion

Resolving a Vector into its Components

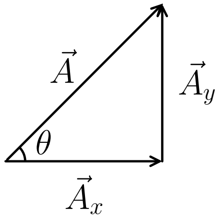
- Any vector can be broken up into its *component vectors*, which are perpendicular to each other.

Suppose $\vec{A}$ is at an angle θ from the positive x-axis.
Its components vectors are $\vec{A}_x$ and $\vec{A}_y$, the magnitudes of which are given by

$$A_x = A\cos\theta$$

$$A_y = A\sin\theta$$

where $A = |\vec{A}|$ is the magnitude of $\vec{A}$ and θ is measured from the positive x-axis.



A vector equals the vector sum of its components:

$$\vec{A} = \vec{A}_x + \vec{A}_y$$

Warning:

$$A \neq A_x + A_y$$

Finding a Vector from its Components

Given components $\vec{A}_x$ and $\vec{A}_y$, we can find the magnitude and direction of vector $\vec{A}$:

Magnitude (Pythagorean Theorem)

$$\left|\vec{A}\right| = \sqrt{A_x^2 + A_y^2}$$

Direction

$$\theta = \tan^{-1}\left(\frac{A_y}{A_x}\right)$$

This definition of θ assumes that $\vec{A}_y$ is the triangle leg opposite to the angle and $\vec{A}_x$ is the triangle leg adjacent to the angle.

Analytical Vector Addition: $\vec{A} + \vec{B} = \vec{R}$

1. **Find components:**

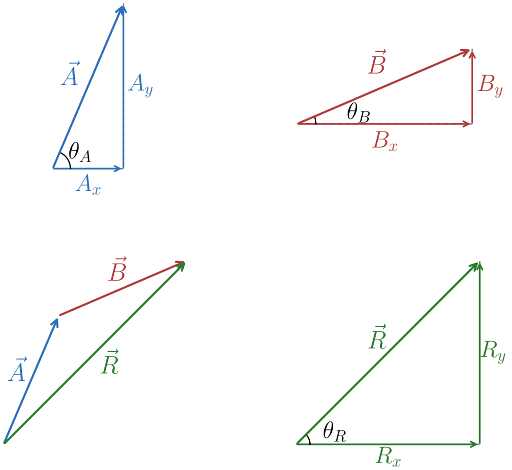
$$A_x = A\cos\theta_A, \quad A_y = A\sin\theta_A$$
$$B_x = B\cos\theta_B, \quad B_y = B\sin\theta_B$$

2. **Sum components:**

$$\vec{R}_x = \vec{A}_x + \vec{B}_x, \qquad \vec{R}_y = \vec{A}_y + \vec{B}_y$$

3. **Find magnitude:** $R = \sqrt{R_x^2 + R_y^2}$

4. **Find direction:**
$\theta = \tan^{-1}(R_y/R_x)$



Example: Analytical Vector Addition

Analytical Vector Addition — (cp2e_ch3_ex3)

Vector $\vec{A}$ has magnitude 53.0 m at 20.0° north of east.
Vector $\vec{B}$ has magnitude 34.0 m at 63.0° north of east.
Find the resultant $\vec{R} = \vec{A} + \vec{B}$.

$$A_x = 53.0 \cos 20.0° = 49.8\,\text{m}, \quad A_y = 53.0 \sin 20.0° = 18.1\,\text{m}$$
$$B_x = 34.0 \cos 63.0° = 15.4\,\text{m}, \quad B_y = 34.0 \sin 63.0° = 30.3\,\text{m}$$

$$R_x = 49.8 + 15.4 = 65.2\,\text{m}, \quad R_y = 18.1 + 30.3 = 48.4\,\text{m}$$

$$R = \sqrt{65.2^2 + 48.4^2} = 81.2\,\text{m}$$

$$\boxed{\theta = \tan^{-1}\left(\frac{48.4}{65.2}\right) = 36.6° \text{ north of east}}$$

Analytical methods give exact answers, unlike graphical methods which depend on the precision of your ruler and protractor.

Check Your Understanding

A vector $\vec{A}$ has magnitude $10.0\,\mathrm{m}$ at $30.0°$ above the horizontal.

(a) Find its horizontal component A_x.
(b) Find its vertical component A_y.

Come to lecture to see the answer.

Part II

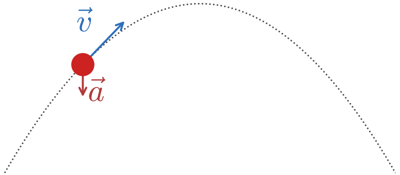
Projectile Motion

Outline of Part II: Projectile Motion

Projectile Motion

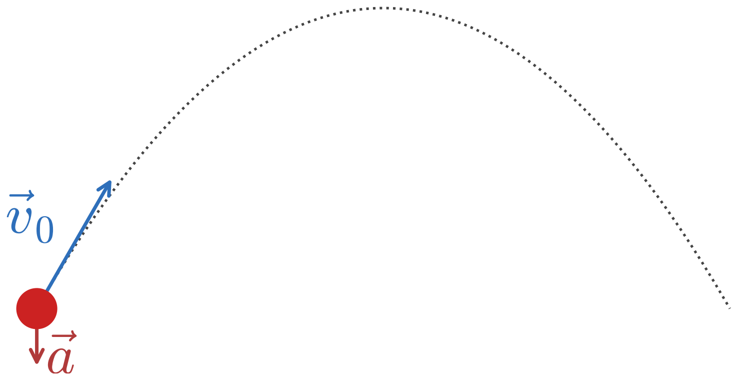
Projectile Motion

- Projectile motion is the motion of an object launched into the air, subject only to the force of gravity.

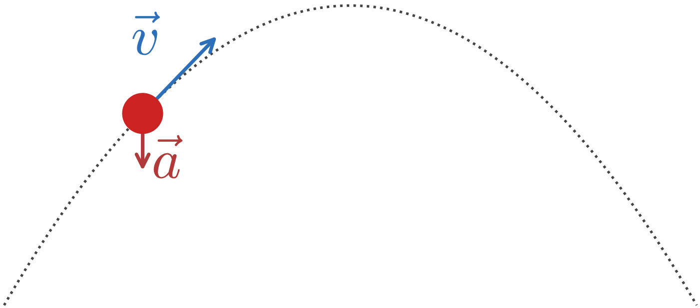


- The path of a projectile is a parabola, called the projectile's trajectory.
- Horizontal: constant velocity $\because a_x = 0 \because$ no horizontal force.
- Vertical: $\vec{a}_y = -g = -9.80\,\frac{\mathrm{m}}{\mathrm{s}^2}$

Illustrating Projectile Motion

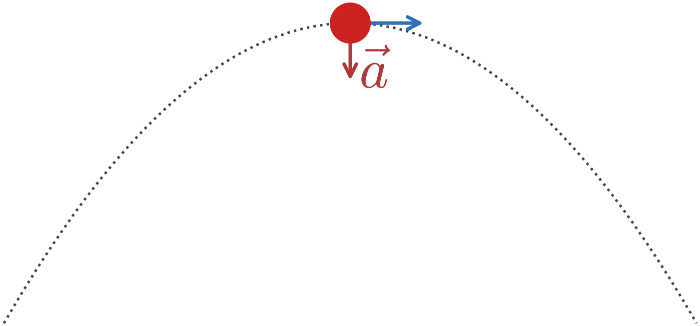


Illustrating Projectile Motion

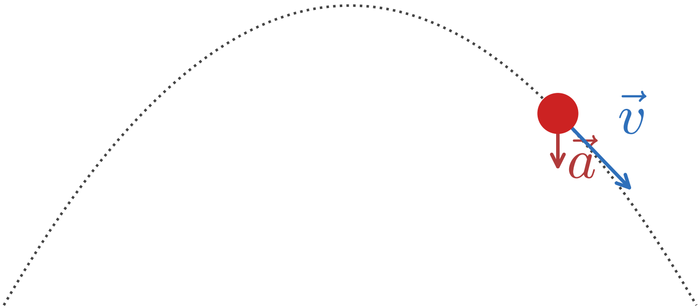


Illustrating Projectile Motion

$$\vec{v} = \vec{v}_x \because \vec{v}_y = 0$$



Illustrating Projectile Motion



Equations Used in Projectile Motion

$$\Delta\vec{x} = \vec{x}_f - \vec{x}_0 \qquad \vec{v}_{\text{avg}} = \frac{\Delta\vec{x}}{\Delta t} \qquad \vec{a}_{\text{avg}} = \frac{\Delta\vec{v}}{\Delta t}$$

$$\vec{v}(t) = \vec{v}_0 + \vec{a}t$$

$$\vec{x}(t) = \vec{x}_0 + \vec{v}_0 t + \frac{1}{2}\vec{a}t^2 \qquad h = \frac{v_{0y}^2}{2g}$$

$$\vec{v}(x)^2 = \vec{v}_0^2 + 2\vec{a}\cdot(\vec{x} - \vec{x}_0)$$

$$\vec{v}_{\text{avg}} = \frac{\vec{v}_f + \vec{v}_0}{2} = \frac{\Delta\vec{x}}{\Delta t} \qquad R = \frac{v_0^2 \sin 2\theta_0}{g}$$

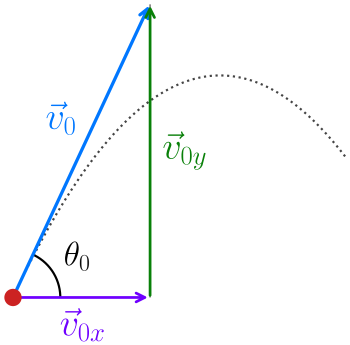
Components of Launch Velocity $\vec{v}_0$

- Suppose a projectile is launched with speed v_0 at angle θ_0 above $+x$-axis.
- The components of $\vec{v}_0$ are then

$$v_{0x} = v_0 \cos\theta_0$$

$$v_{0y} = v_0 \sin\theta_0$$

where v_{0x} is the initial horizontal velocity, v_{0y} is the initial vertical velocity, v_0 is the launch speed, and θ_0 is the launch angle above the horizontal.



- Once the projectile is in the air, v_{0x} remains constant ($\because \vec{a}_x = 0$).
- Only the vertical component changes ($\because \vec{a}_y = -g$) due to gravity.

Projectile Motion — Examining a Projectile's Horizontal Motion

Horizontal velocity is constant $\left(\because \Delta\vec{v}_x = 0 \because \vec{a}_x = 0 \because \vec{F}_{net,x} = 0\right)$. Therefore,

$$v_x(t) = v_{0x}$$

$$\vec{x}(t) = \vec{x}_0 + \vec{v}_{0x}\,t$$

where $\vec{x}$ is the horizontal position, $\vec{x}_0$ is the initial horizontal position, $\vec{v}_{0x}$ is the constant horizontal velocity, and t is time.

The horizontal motion of a projectile is identical to uniform motion (constant velocity, zero acceleration) in the x-direction.

Projectile Motion — Examining a Projectile's Vertical Motion

The vertical motion is identical to free fall:

$$\vec{v}_y(t) = \vec{v}_{0y} + \vec{a}_y t$$

$$\vec{y}(t) = \vec{y}_0 + \vec{v}_{0y} t + \tfrac{1}{2}\vec{a}_y t^2$$

$$v_y^2 = v_{0y}^2 + 2\vec{a}_y \cdot (\vec{y} - \vec{y}_0)$$

where $\vec{y}$ is the vertical position, $\vec{v}_y$ is the vertical velocity, $\vec{v}_{0y}$ is the initial vertical velocity, $\vec{a}_y = -g = -9.80\,\frac{\mathrm{m}}{\mathrm{s}^2}$, and t is time.

$$\vec{v}_y(t) = \vec{v}_{0y} - gt$$

$$\vec{y}(t) = \vec{y}_0 + \vec{v}_{0y} t - \tfrac{1}{2} g t^2$$

$$v_y^2 = v_{0y}^2 - 2g(y - y_0)$$

Time t is common to all motion $\therefore$ we use it to link the x-direction kinematic equations and the y-direction kinematic equations.

How to Solve Projectile-Motion Problems

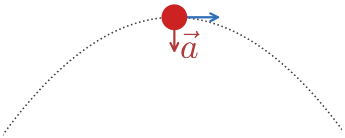
- Treat horizontal and vertical motions as two separate 1-D problems.
- Use time t as the linking variable connecting both directions.
- Combine components using vector addition if needed:

$$v = \sqrt{v_x^2 + v_y^2}, \qquad \theta_v = \tan^{-1}\left(\frac{v_y}{v_x}\right)$$

💡 Set $\vec{x}_0 = 0$ and $\vec{y}_0 = 0$ at the launch point to simplify algebra.

Velocity in the y-direction is Zero at the Peak

$$\vec{v} = \vec{v}_x \because \vec{v}_y = 0$$



- At the projectile's maximum height (the peak), $\vec{v}_y = 0$.

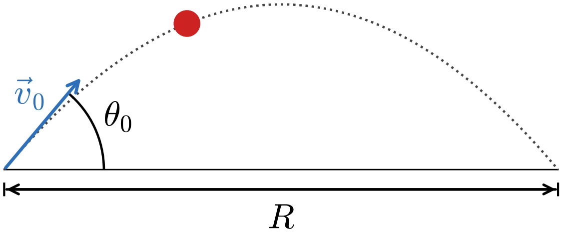
Using $v_y^2 = v_{0y}^2 + 2\vec{a}_y \cdot (\vec{y} - \vec{y}_0) \quad \longrightarrow \quad v_y^2 = v_{0y}^2 - 2gh$:

$$h = \frac{v_{0y}^2}{2g} = \frac{(v_0 \sin\theta_0)^2}{2g}$$

where h is the maximum height above the launch point, v_0 is the launch speed, θ_0 is the launch angle, and $g = 9.80\,\frac{\mathrm{m}}{\mathrm{s}^2}$.

Maximum height depends only on the vertical component of the initial velocity.

Horizontal Range R



- Maximum range occurs at $\theta_0 = 45°$
- Complementary angles (e.g., 30° and 60°) give equal ranges

The **range** is the total horizontal distance traveled when the projectile lands at the same height from which it was launched:

$$R = \frac{v_0^2 \sin 2\theta_0}{g}$$

where R is the range, v_0 is the launch speed, θ_0 is the launch angle, and $g = 9.80\,\frac{\text{m}}{\text{s}^2}$.

📝 R is measured with the projectile landing at the same height! $(\vec{y}_f = \vec{y}_0)$

🔬 Due to the effect of air resistance, the true optimal angle drops to $\approx 38°$ and ranges are reduced.

Example: Fireworks Projectile

Fireworks Shell — (cp2e_ch3_ex4)

A fireworks shell is launched with $v_0 = 70.0\,\frac{\text{m}}{\text{s}}$ at $\theta_0 = 75.0°$ above horizontal. Find (a) the maximum height and (b) the horizontal distance at maximum height.

$$v_{0x} = 70.0\cos 75.0° = 18.1\,\frac{\text{m}}{\text{s}}$$
$$v_{0y} = 70.0\sin 75.0° = 67.6\,\frac{\text{m}}{\text{s}}$$

$$\text{(a) } h = \frac{v_{0y}^2}{2g} = \frac{(67.6\,\frac{\text{m}}{\text{s}})^2}{2(9.8\,\frac{\text{m}}{\text{s}^2})}$$

$$\boxed{h = 233\,\text{m}}$$

$$\text{(b) } \vec{v}_y(t) = \vec{v}_{0y} + \vec{a}_y t = \vec{v}_{0y} - gt$$
$$\vec{v}_y(t_p) = 0 = v_{0y} - gt_p \therefore t_p = \frac{v_{0y}}{g}$$
$$\vec{x}(t) = \vec{x}_0 + \vec{v}_{0x}t + \tfrac{1}{2}\vec{a}_x t^2 = \vec{v}_{0x}t$$
$$x(t_p) = x_p = v_{0x}t_p = v_{0x}\left(\frac{v_{0y}}{g}\right) = 18.1\,\frac{\text{m}}{\text{s}}\left(\frac{67.6\,\frac{\text{m}}{\text{s}}}{9.8\,\frac{\text{m}}{\text{s}^2}}\right)$$

$$\boxed{x_p = 125\,\text{m}}$$

At maximum height, $\vec{v}_y = 0$ and $\vec{v}_x = \vec{v}_{0x} = 18.1\,\frac{\text{m}}{\text{s}}$.

Time to peak $t_p = t_{peak}$.

Find horizontal distance to the peak (x_p) by using $t = t_p \because x(t_p) = x_p$

Example: Projectile Launched from a Height

Hot Rock — (cp2e_ch3_ex5)

A hot rock is launched at $v_0 = 25.0\,\frac{\text{m}}{\text{s}}$ and $\theta_0 = 35.0°$ above the horizontal.
It lands 20.0 m below the launch point.
Find (a) the time of flight and (b) the impact speed and angle.

$$v_{0x} = v_0 \cos\theta_0, \;\; v_{0y} = v_0 \sin\theta_0, \;\; \vec{y}_f = -20.0\,\text{m}$$

(a) $$\vec{y}_f = \vec{y}_0 + \vec{v}_{0y}\, t + \tfrac{1}{2}\vec{a}_y\, t^2$$

$$\boxed{t = 3.96\,\text{s}} \quad \text{(from quadratic equation)}$$

(b) $$\vec{v}_{fy} = \vec{v}_{0y} + \vec{a}_y\, t = 14.3\,\tfrac{\text{m}}{\text{s}} - 9.8\,\tfrac{\text{m}}{\text{s}^2}(3.96\,\text{s})$$

$$\vec{v}_{fy} = -24.5\,\tfrac{\text{m}}{\text{s}}. \quad \theta_f = \arctan\frac{v_{fy}}{v_{fx}} = 50.1°$$

$$v_f = \sqrt{v_{fx}^2 + v_{fy}^2} = \sqrt{(20.5\,\tfrac{\text{m}}{\text{s}})^2 + (24.5\,\tfrac{\text{m}}{\text{s}})^2}$$

$$\boxed{\vec{v}_f = 31.9\,\tfrac{\text{m}}{\text{s}} \text{ at } 50.1° \text{ below horizontal}}$$

The quadratic,
$-y_f = v_{0y}t - \frac{1}{2}gt^2 = -20.0$,
yields a positive root and a negative root. $\because t > 0$, only keep the positive root.

- $\vec{v}_{fy} < 0$ because the rock is moving downward at impact.
- $\vec{v}_x$ is constant throughout the flight.

Check Your Understanding

A ball is thrown horizontally (i.e., $\theta_0 = 0°$) from a cliff at $15.0\,\frac{\text{m}}{\text{s}}$.

(a) What is v_{0y}?

(b) Does the ball have horizontal acceleration during flight?

(c) What pulls the ball downward?

Come to lecture to see the answer.

Summary: Projectile Motion

- Horizontal velocity is **constant** throughout the flight.
- Vertical motion is **identical to free fall**.
- Time t is the **shared variable** linking horizontal and vertical equations.
- Range is maximized at $\theta_0 = 45°$ (in vacuum).
- Complementary angles give **equal range** (assuming no air resistance).
- Projectile motion assumes **negligible air resistance**; real trajectories differ.

Part III

Addition of Velocities

Outline of Part III: Addition of Velocities

Relative Velocity

- **Relative velocity** is the velocity of an object as measured from a specific **reference frame**.
- The same object can have **different velocities** relative to different observers.

- A passenger walking forward in a moving train has one velocity relative to the train, and a different (greater) velocity relative to the ground.
- A boat crossing a river has a velocity relative to the water and a different velocity relative to the riverbank.

📝 There is technically no such thing as "absolute" velocity — velocity is *always* measured relative to a reference frame.

Vector Addition of Velocities (General 2-D Vector Addition)

When an object A moves with another object B, the velocities add as vectors:

$$\vec{v}_{B,A} = \vec{v}_B' - \vec{v}_A'$$
$$\vec{v}_{B,A} = \vec{v}_B' + (-\vec{v}_A')$$

where A and B are objects and $\vec{v}_{B,A}$ is the velocity of B relative to A, $\vec{v}_B'$ is the velocity of B relative to the ground, and $\vec{v}_A'$ is the velocity of A relative to the ground.

You can intuit this equation yourself!

Suppose Bob drives 70 $\frac{\text{mi}}{\text{h}}$ eastward and Alice drives 50 $\frac{\text{mi}}{\text{h}}$ eastward.
As Bob passes Alice, what does she perceive his velocity to be?

$$\vec{v}_{B,A} = \vec{v}_B' - \vec{v}_A'$$
$$\vec{v}_{B,A} = 70\ \frac{\text{mi}}{\text{h}} - 50\ \frac{\text{mi}}{\text{h}} = \boxed{20\ \frac{\text{mi}}{\text{h}}}$$

What does she perceive to be the velocity of her coffee in her cup holder?

$$\vec{v}_{C,A} = \vec{v}_C' - \vec{v}_A'$$
$$\vec{v}_{C,A} = 50\ \frac{\text{mi}}{\text{h}} - 50\ \frac{\text{mi}}{\text{h}} = \boxed{0\ \frac{\text{mi}}{\text{h}}}$$

Velocities are vectors $\therefore$ add their components, not their magnitudes.

Example: Boat Crossing a River

Boat and River Current — (cp2e_ch3_ex6)

A boat moves perpendicular to a riverbank at $0.75\ \frac{\text{m}}{\text{s}}$. The river current flows at $1.20\ \frac{\text{m}}{\text{s}}$ parallel to the bank. (Note: the bank is the stationary ground.)
Find (a) the boat's speed relative to the bank and (b) its direction of travel.

(a) $$|\vec{v}'_B| = \sqrt{|\vec{v}'_{Bx}|^2 + |\vec{v}'_{By}|^2}$$

$$\vec{v}_{B,Ax} = \vec{v}'_{Bx} - \vec{v}'_{Ax}$$
$$\vec{v}'_{Bx} = \vec{v}_{B,Ax} + \vec{v}'_{Ax}$$
$$\vec{v}'_{Bx} = 0 + 1.20\ \frac{\text{m}}{\text{s}} = 1.20\ \frac{\text{m}}{\text{s}}$$

$$\vec{v}'_{By} = 0.75\ \frac{\text{m}}{\text{s}}$$

$$|\vec{v}'_B| = \sqrt{1.20\ \frac{\text{m}}{\text{s}}^2 + 0.75\ \frac{\text{m}}{\text{s}}^2}$$
$$\boxed{v'_B = 1.4\ \frac{\text{m}}{\text{s}}}$$

River (Obj. A), Boat (Obj. B)

$$\vec{v}_{B,A} = \vec{v}'_B - \vec{v}'_A \quad \text{also means:}$$

$$\vec{v}_{B,Ax} = \vec{v}'_{Bx} - \vec{v}'_{Ax}$$
$$\vec{v}_{B,Ay} = \vec{v}'_{By} - \vec{v}'_{Ay}$$

- The current carries the boat downstream even though the boat aims straight across.

Example: Boat Crossing a River

Boat and River Current — (cp2e_ch3_ex6)

A boat moves perpendicular to a riverbank at $0.75\,\frac{\text{m}}{\text{s}}$. The river current flows at $1.20\,\frac{\text{m}}{\text{s}}$ parallel to the bank. (Note: the bank is the stationary ground.)
Find (a) the boat's speed relative to the bank and (b) its direction of travel.

(b) $\theta_B = \tan^{-1}\left(\frac{v'_{By}}{v'_{Bx}}\right)$

$$\theta_B = \tan^{-1}\left(\frac{0.75\,\frac{\text{m}}{\text{s}}}{1.20\,\frac{\text{m}}{\text{s}}}\right)$$

$$\boxed{\theta_B = 32.0^\circ \text{ from } +x \text{ direction}}$$

River (Obj. A), Boat (Obj. B)

$\vec{v}_{B,A} = \vec{v}'_B - \vec{v}'_A$ also means:

$$\vec{v}_{B,Ax} = \vec{v}'_{Bx} - \vec{v}'_{Ax}$$
$$\vec{v}_{B,Ay} = \vec{v}'_{By} - \vec{v}'_{Ay}$$

- The current carries the boat downstream even though the boat aims straight across.

Validity of Standard Vector Addition: Classical Relativity

Classical relativity applies when all speeds are much less than the speed of light ($c \approx 3 \times 10^8\ \frac{\text{m}}{\text{s}}$).

At these everyday speeds, velocities add by standard **vector addition**.

Coin Dropped in a Moving Plane

- Plane speed: $260\ \frac{\text{m}}{\text{s}}$ (horizontal)
- Coin drops $1.50\,\text{m}$ (vertical)
- Relative to plane: coin falls straight down at $5.42\ \frac{\text{m}}{\text{s}}$
- Relative to Earth: coin moves at $\approx 260\ \frac{\text{m}}{\text{s}}$ nearly horizontally

The same physical motion appears differently to observers in different reference frames.

Einstein's special relativity modifies the classical addition of velocities when speeds approach c.

Check Your Understanding

A train moves due north at $45.0\,\frac{\mathrm{m}}{\mathrm{s}}$ relative to the ground. A bird flies due west at $20.0\,\frac{\mathrm{m}}{\mathrm{s}}$ relative to the ground.

What is the train's speed relative to the bird?

Come to lecture to see the answer.

Key Takeaways: Addition of Velocities

- Velocity is **always relative** to a reference frame — there is no absolute velocity.
- Velocities in two dimensions add as **vectors**, not scalars.
- The same physical motion looks different to observers in different reference frames.
- Choosing coordinates aligned with a known velocity simplifies calculations.
- Classical (Galilean) velocity addition applies when $v \ll c$.

Next Lecture

Chapter 4: Dynamics — Force and Newton's Laws of Motion

- So far, we have described **how** objects move (kinematics).
- Next, we ask **why** objects move: what forces cause or change motion?
- We will introduce Newton's three laws and apply them to a variety of situations.

- The tools developed in Chapters 2 and 3 (1-D and 2-D kinematics) will continue to be used throughout the course.
- Make sure you are comfortable with vector components and the kinematic equations.